

Large Language Models: A Survey of Surveys

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Not only did the growing interest in Large Language Models (LLMs) lead to a multitude of applications, news articles, social media posts, and new products, but it also resulted in a significant increase in research publications. To gain a better understanding of this vast number of publications, surveys help provide practitioners and researchers with much-needed overviews.

However, we have reached a point with tens of thousands of LLM publications, and the number of surveys on LLM publications has grown into hundreds. Ironically, the same surveys that set out to bring order and structure now contribute to the convolution of the space. For example, someone interested in the use of LLMs for the health sector has more than 80 potential surveys to choose from. To address this challenge, we carry out a tertiary literature review to gather and analyze LLM-related surveys, reviews, and mapping studies. By doing so, we aim to help practitioners and researchers navigate the vast array of existing surveys.

In total, we found 424 LLM surveys that have been published up to September 2024 that are included in this study. We devise a taxonomy and categorise surveys according to their main focus (e.g., fine-tuning of LLMs, application for software engineering tasks). To further support the navigation of LLM surveys and keep up to date, we created a GitHub repository that extends our scope to a total of 984 publications published up to August 2025, which is available from <https://github.com/dataSED-condense/LLM-Survey-Survey>.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Natural language processing**.

Additional Key Words and Phrases: large language model, literature survey, tertiary study

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1 Introduction

Summaries are useful tools for providing overviews that help facilitate the understanding of diverse topics. In research, summaries are typically conducted through *secondary studies* (e.g., survey, mapping study, literature review) [83, 126]. Going one step further, *tertiary reviews* systematically analyze and summarize secondary studies [148]. Such tertiary reviews have proven useful across various domains, such as software engineering [43, 83], medical deep learning [62], economics [46], machine learning [150], requirements engineering [12], and sentiment analysis [181].

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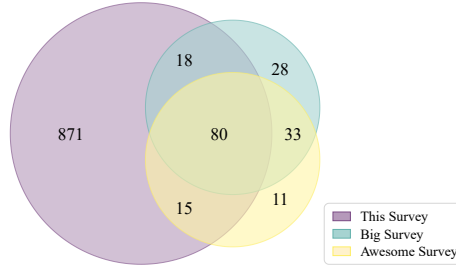


Fig. 1. Venn diagram illustrating the overlap between surveys from our systematic search and two repositories.

The surging popularity of Large Language Models (LLMs) over the recent years has led to thousands of research articles and, in turn, hundreds of secondary studies. This volume of secondary studies creates a need for a structured overview, and we argue that it is time to conduct a tertiary study of the field of LLMs. By creating an overview, we support researchers and practitioners in navigating the field of surveys, finding relevant ones when learning about LLMs, and understanding which aspects have been studied when designing new surveys.

To the best of our knowledge, no tertiary study on LLMs has been published. However, we are aware of two GitHub repositories that provide a list of LLMs surveys. These are: ABigSurvey-OfLLMs¹ and Awesome-LLM-Survey,² with 159 and 139 listed surveys on LLMs respectively. While valuable resources, these surveys did not follow a systematic search methodology. We extend far beyond their scope by carrying out a systematic literature search and collecting a total of 424 surveys that are included in this article, as well as an additional 560 that are included in our GitHub repository (see Section 2 for more details). The overlap of these two repositories and our collected surveys is shown in Figure 1, highlighting the additional studies collected.

Figure 2 shows the structure of our survey and how we categorize the existing surveys.

2 Survey Methodology and Search Results

2.1 Search Procedure

To search the literature for secondary studies about LLMs, we make use of the publication database dblp.³ dblp contains publications from more than 1,800 journals and 6,000 conferences from the computer science domain, as well as non-peer-reviewed papers from arXiv. In particular, we use dblp to carry out a search of relevant publications based on filtering their titles.

To ensure that we obtain relevant search results, we define two sets of keywords. The first set contains terms related to language models, while the second contains terms related to literature collection (inspired by Kotti et al. [150]):

- LLM keywords: LLM, Language Model.
- Survey keywords: Survey, Overview, Literature, Review, Background, Research, Taxonomy, Systematic.

In addition to requiring that publication titles contain both keyword types, we treat them as inclusion criteria for our paper collection:

- (1) LLMs: The paper focuses on language models.
- (2) Literature overview: The paper represents a secondary study by collecting and presenting other works.

¹ <https://github.com/NiuTrans/ABigSurveyOfLLMs>, last updated on 19th February 2025

² <https://github.com/HqWu-HITCS/Awesome-LLM-Survey>, last updated on 25th of May 2025

³ <https://dblp.org>

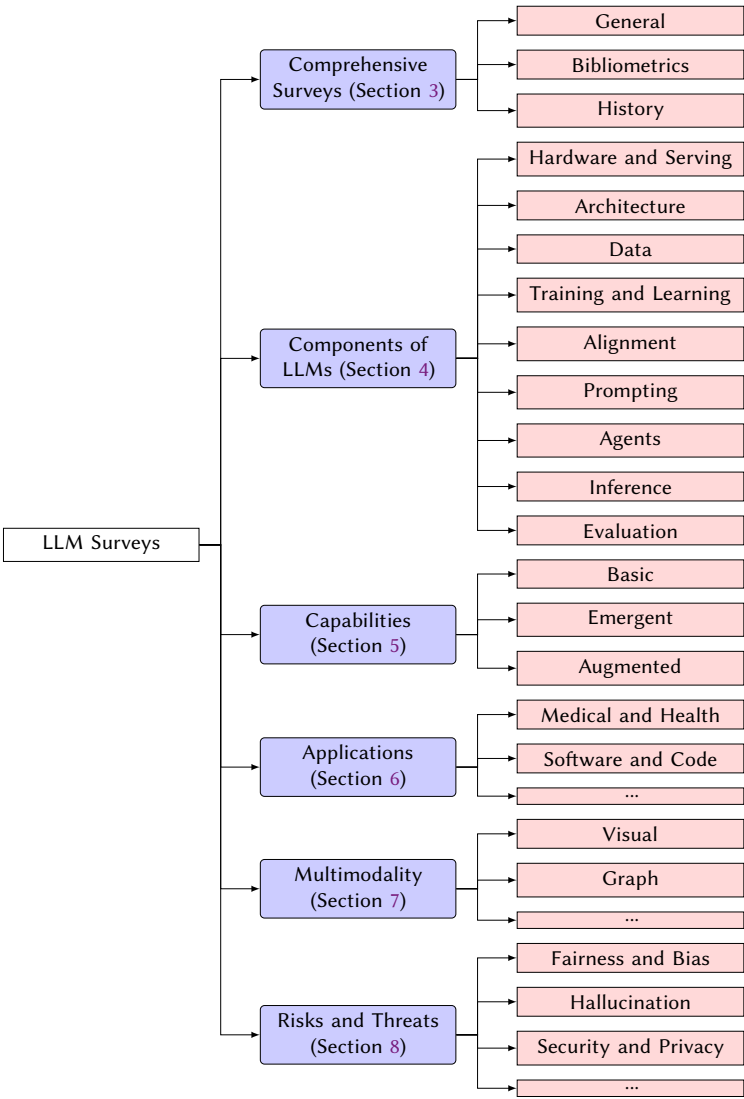


Fig. 2. Structure of this study.

We exclude all studies that do not match these criteria, and omit studies that are not written in English. We check inclusion in two stages. First, we determine relevance of the search results based on their title. For instance, this removes literature on the study of “language modeling”. Second, we read each paper with a suitable title and make a final inclusion decision based on its content.

2.2 Selection and Search Results

Table 1 summarizes the results of our search, which we carried out on 10th of September 2024. We start with a total of 1,173 unique publications from dblp, which fit at least one of the keyword combinations. 461 of these agree with our inclusion criteria according to their titles. After examining the 461 papers, we exclude 37 and end up with a total of 424 studies that are included in our survey and presented in the following sections. To ensure the timeliness of our work, we carried out an

Table 1. Summary of search results. The search was carried out on the 10th of September 2024 on dblp. Results for the updated search carried out on the 15th of August 2025 are shown in (blue). The additional 560 studies can be found in our GitHub repository.

Survey Keyword	LLM Keyword		# Papers		
	“LLM”	“Language Model”	Unique	Title	Content
“Survey”	63 (+181)	427 (+476)			
“Overview”	6 (+12)	43 (+35)			
“Literature”	18 (+54)	65 (+91)			
“Review”	43 (+146)	231 (+291)	1,173	461	424
“Background”	0 (+2)	8 (+9)	(+1,672)		(+560)
“Research”	33 (+146)	201 (+241)			
“Taxonomy”	14 (+40)	48 (+51)			
“Systematic”	25 (+87)	134 (+169)			

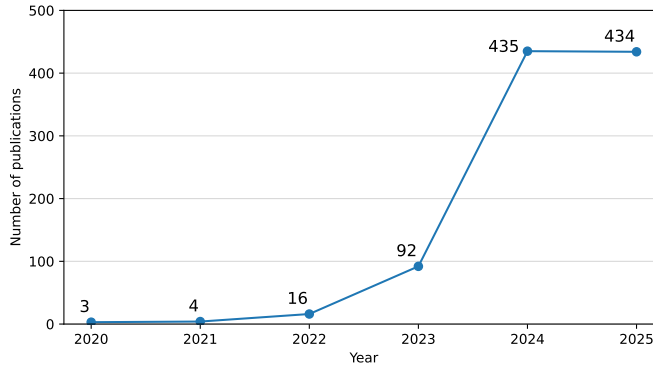


Fig. 3. Number of publications per year. The count for 2025 is based on a cut-off date of 15th of August 2025.

identical search on the 15th of August 2025, to find surveys that have been published in the last year. This resulted in an additional 560 surveys. Due to the large number of recent publications, we decided to only include them in our supplementary online repository.⁴ The temporal distribution of these publications is shown in Figure 3.

3 Overviews and Comprehensive Surveys

We start the presentation of existing surveys by presenting general overviews that are helpful as an introduction to learn about LLMs. These surveys stand out by their comprehensiveness and the consideration of several aspects of LLMs. In addition to comprehensive surveys, we include surveys about bibliometrics and the history of LLMs to provide extensive background details.

Table 1 shows the included surveys for this section. For each, we list basic information (Authors, venue, year of publication), as well as the number of references they have and how often they have been cited. These can be useful indicators for their comprehensiveness and popularity. Lastly, we give a Unique Selling Point (USP), a point of focus which differentiates them from other surveys.

3.1 Comprehensive

Zhao et al. [417] created the most comprehensive survey on LLMs to date. This can not only be seen by the high number of references included (946), but also its popularity (over 5000 citations). This

⁴ <https://github.com/dataSED-condense/LLM-Survey-Survey>

Table 2. Overview of comprehensive surveys and their unique selling point. The number of citations was collected from Google Scholar on the 21st of August 2025. The # studies shows how many publications are covered by the respective surveys.

Authors	Venue	Year	# Studies	Citations	Focus	USP
Movva et al. [222]	NAACL	2024	59	24	Bibliometrics	Industry and Academia (roles)
Fan et al. [65]	arXiv	2023	86	199	Bibliometrics	Research topics
Naveed et al. [223]	arXiv	2023	487	1496	Comprehensive	Architecture details
Raiaan et al. [254]	IEEE Access	2024	187	667	Comprehensive	Datasets per model
Zhao et al. [417]	arXiv	2023	946	5658	Comprehensive	Detailed settings
Yang et al. [368]	ACM TKDD	2024	143	1212	Comprehensive	NLP tasks
Minaee et al. [218]	arXiv	2024	243	1263	Comprehensive	Capabilities
Liu et al. [196]	arXiv	2024	175	144	Comprehensive	Training & Inference
Ling et al. [185]	arXiv	2024	297	57	Comprehensive	Specialization
Guo and Yu [93]	arXiv	2022	175	34	Comprehensive	Domain Adaptation
Wang et al. [317]	arXiv	2024	305	35	Comprehensive	Challenges and Opportunities
Miao et al. [216]	arXiv	2023	375	103	Comprehensive	Systems and Serving
Wei et al. [330]	arXiv	2023	223	74	History	Conventional models and linguistic units
Chu et al. [38]	arXiv	2024	88	93	History	Advancement of LLMs
Kumar [153]	Artif. Intell. Rev.	2024	249	138	History	Word embeddings, Deep Learning

scale allows the survey to cover all important aspects of LLMs and is the only survey to consider aspects such as “scaling laws”, which have not been covered by the other surveys.

There are several other surveys that provide a comprehensive overview of LLMs. While some of their contents naturally overlap, we outline their unique viewpoints. Raiaan et al. [254] provided an overview of the different sources for datasets (e.g., webpages, books, code). Naveed et al. [223] listed details on the architecture of LLMs. This includes information such as training objective, vocabulary size, type of attention, number of layers, attention heads, and hidden states. Minaee et al. [218] provided an overview of the capabilities of language models. Moreover, they survey the components necessary for building LLMs. Miao et al. [216] covered the serving of LLMs and optimization for faster inference time via modifying the models themselves or the hosting system.

Yang et al. [368] include the most comprehensive description of NLP tasks for LLMs. The survey by Liu et al. [196] focused on training and inference, ranging from the data processing stage to different fine-tuning paradigms and methods for speeding up the inference. Ling et al. [185] addressed the adaptation of LLMs to different domains in their survey. These techniques range from augmentation with external knowledge to fine-tuning. Similarly, Guo and Yu [93] described domain adaptation via data augmentation, model optimization (training) and model personalization.

3.2 Bibliometric

Fan et al. [65] carried out a bibliometric study covering 5752 publications from the Web of Science (WoS) Core Collection, collected from 2017 to early 2023. They investigated topics addressed by these publications and divided them into five categories: algorithm and NLP tasks, medical and engineering applications, social and humanitarian applications, critical studies, and infrastructure. Among these, “Algorithm and NLP tasks” span the majority of publications (54%), while “Infrastructure” and “Critical studies” cover less than 2% each. The countries which produced the highest number of research in this period are China and the USA. In terms of the collaboration among institutes, USA and UK have the highest centrality score.

Movva et al. [222] performed a study to reveal the influence of LLMs on AI research, and analyzed 16,979 LLM-related papers from arXiv during the period of January 2018 to September of 2023. They observed that many authors have not previously published NLP-related research, and a growing interest on the societal impact of LLMs. Similar to the findings by Fan et al. [65], US and China-based

institutes contributed the highest number of publications. Overall, Movva et al. [222] observed few collaborations across countries.

3.3 History

Another three surveys outline current advances in language models while providing information on the history and early approaches [38, 153, 330]. For instance, Wei et al. [330] started their survey with an overview on conventional language models (e.g., structural and bidirectional language models), while also describing various linguistic units (i.e., characters, words, subwords, phrases, sentences). The role of word embeddings and deep learning for language models is addressed by Kumar [153]. Chu et al. [38] considered approaches ranging from 1990 (statistical language models) to 2023 (large language models).

4 Components of LLMs

This section outlines surveys that addressed the different components required for the training and use of LLMs. We structure our review around nine key components as shown in Figure 4. This taxonomy is inspired by the work of Naveed et al. [223] and Minaee et al. [218].

4.1 Hardware and Serving

LLMs are compute-intensive machine learning models and therefore require a certain degree of compute power and hardware infrastructure to be used. For instance, the running of LLMs can benefit from the use of GPUs [305] and high-performance computing [31]. The efficiency of training and applying LLMs has been improved from a diverse range of components [138, 308], such as processing units, storage systems, scheduling, and memory management [55, 61, 163, 305, 356, 386, 428]. In addition, there are two dedicated surveys on improving efficiency via the key-value (KV) cache [274] (during training and inference) and compute-in-memory (i.e., reduces overhead of memory access by performing computations in memory) [334].

A frequently mentioned approach for accelerating the training process is parallelization [9, 18, 55, 61, 305]. Here, Duan et al. [61] mentioned three different types (Hybrid, Auto, Heterogeneous) and descriptions on optimizing communication. Another hardware consideration is the device on which LLMs are run. These can be edge devices [14, 250, 356] or in the cloud [9, 163, 260, 356, 386, 428].

4.2 Architecture

In this section, we present surveys that describe existing model types and information on their architectures. For instance, Gao et al. [79] listed models and provided details, such as their number of parameters and underlying base models. In addition, they evaluated 32 of them in various settings (e.g., zero-shot, few-shot, multi-modal) and presented tools that support the development with and for LLMs. Pahune and Chandrasekharan [229] showed the different available versions for each of the models and hardware details for their implementation.

Other surveys focus on specific model families. Kukreja et al. [152] considered open-source models, with particular focus on FALCON, BLOOM, and Llama2, for which data collection, architecture, and training stages are described. Kalyan [139] focused on GPT language models, in particular models ranging from GPT-3 to GPT-4, and collected their application to downstream tasks (e.g., text classification, information extraction, coding). Alipour et al. [5] focused on ChatGPT and OpenAI (e.g., the OpenAI playground). Other models were introduced as alternatives to ChatGPT. Lu et al. [201] considered different methods for LLM collaboration. For instance, LLM responses can be merged, or one can create an ensemble of multiple LLMs.

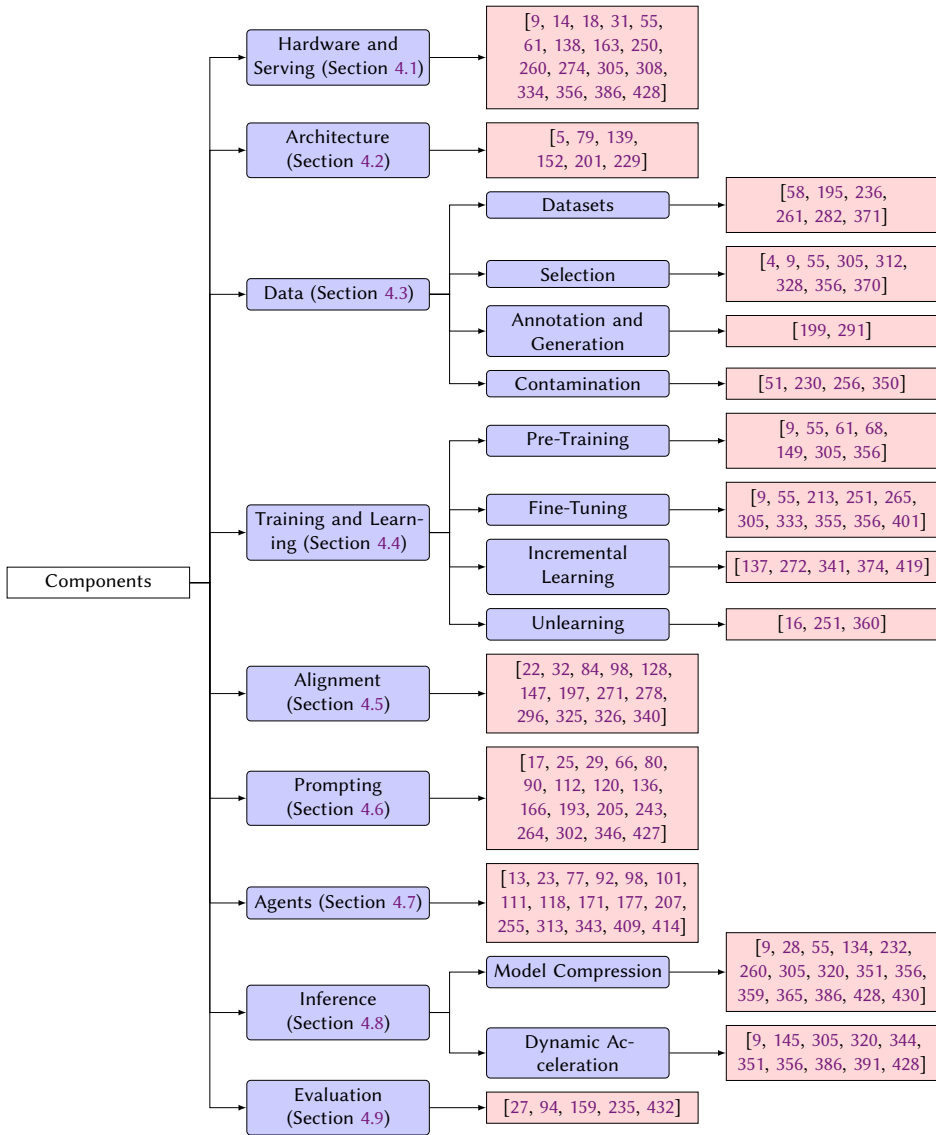


Fig. 4. Taxonomy of surveys on LLM components.

4.3 Data

The characteristics of an LLM are fundamentally determined by the data used in its creation and evaluation. Consequently, surveys in this field explore the entire data lifecycle, from the composition of datasets to the evaluation of their quality.

Datasets: Liu et al. [195] presented an exhaustive overview of datasets for large language models. They considered a total of 444 datasets from five categories: pre-training, instruction fine-tuning, preference, evaluation, and NLP. Srivastava and Memon [282] presented 52 datasets for the open-domain question-answering tasks, and the study by Yang et al. [371] reviewed datasets for causal reasoning benchmarks. Röttger et al. [261] presented 102 datasets for safety evaluation.

While large datasets can be beneficial for LLM performance, one needs to be careful when obtaining data from public sources. Challenges faced when using web-mined corpora for pre-training LLMs have been reviewed by Perelkiewicz and Poswiata [236]. Among others, they presented challenges based on sensitive information, bias or the low quality of data. Du et al. [58] gathered 32 datasets (16 for pre-training and 16 for fine-tuning) while focusing on their quality and quantity.

Data selection: Datasets for training LLMs contain an enormous amount of samples with varying characteristics. While all data samples can be used for training, one can also select the ones most suitable for one's goal. For this purpose, the amount of training data can be reduced via deduplication, sampling or selection [9, 55, 305, 356]. Albalak et al. [4] surveyed data selection for LLMs. Methods are organized based on the type of data (e.g., data selection for pre-training, in-context learning). Additionally, they provided an overview of the main objectives of data selection at each stage of the training process (e.g., the main objective of data selection for fine-tuning is bias reduction and model performance). Wang et al. [312] specialized in selecting data for instruction tuning.

Wang et al. [328] considered data collection from a data management perspective. This includes concerns regarding data quality and quantity (e.g., filtering strategies) for pre-training and fine-tuning datasets. Lastly, [370] surveyed the impact of adding source code to the training data of LLMs, and found that it can improve downstream performance.

Data annotation and generation: While datasets for training LLMs are often obtained from human-created sources, LLMs themselves can be used to augment or enhance existing data, be it by generating new data from scratch (data generation) or providing additional information to existing data (data annotation). For instance, Long et al. [199] surveyed synthetic data generation with LLMs to outline the workflow for data generation, consisting of generation, curation, and evaluation of synthetic data. Tan et al. [291] considered different facets of the data annotation process with LLMs (generation, assessment and utilization).

Data contamination: Data contamination is a problem that arises when the training data of LLMs overlap with the evaluation benchmarks. We found four surveys summarizing approaches for detection and mitigation of data contamination. Palavalli et al. [230] considered two severities of data contamination (instance level, dataset level) and examined them in two case studies (i.e., summarization, question answering). Xu et al. [350] considered the severity of data contamination (i.e., semantic, information, data, label level) and presented several tasks where contamination has been observed (e.g., code generation, sentiment analysis). Deng et al. [51] considered language model types (white-box, gray-box, black-box LLMs) when it comes to data contamination as well as several methods for detecting data contamination. Lastly, Ravaut et al. [256] organized contamination detection approaches based on open-data (dataset is known) and closed-data (dataset is not known).

4.4 Training and Learning

By leveraging large amounts of data, LLMs learn patterns that shape their performance across different stages. From initial pre-training to continuous adaptation, these stages allow them to acquire and refine their capabilities.

Pre-train: Pre-training describes the initial training stage of LLMs, in which models learn a general understanding of texts and language. Kotei and Thirunavukarasu [149] surveyed different pre-training techniques (from scratch, incessant pretraining, based on knowledge inheritance, multi-task pre-training). Afterwards, they discussed how this knowledge can be transferred to downstream tasks via fine-tuning. Fang et al. [68] reviewed metrics to consider for the training process and monitoring of the training success. While we found no other dedicated studies, several pre-training techniques have been covered by comprehensive surveys. For instance, the most

frequently considered method for improving the efficiency of the pre-training process is mixed-precision training [9, 55, 61, 305, 356].

Fine-Tune: After pre-training, LLMs can be fine-tuned for specific tasks, which usually involves smaller datasets of higher quality. Weng [333] considered several fine-tuning paradigms, such as multi-task learning, knowledge distillation, transfer learning, and few-shot learning. Other surveys considered specific learning paradigms, such as federated learning [251], multi-task learning [265], or instruction-tuning [401]. A larger subset of surveys addressed the efficiency of the fine-tuning process via Parameter Efficient Fine-Tuning (PEFT) [9, 55, 305, 355, 356].

Xu et al. [355] covered the efficiency of the training of LLMs by PEFT methods. Rather than tuning the entire model (all parameters), a limited subset is fine-tuned to save time and memory. They categorized PEFT methods into 5 types: additive fine-tuning, partial fine-tuning, reparameterized fine-tuning, hybrid fine-tuning, and unified fine-tuning. In addition to the collection and description of a multitude of PEFT methods, Xu et al. carried out an empirical comparison of fine-tuning a RoBERTa model and 11 PEFT methods. Another PEFT method that received a survey of its own is LoRA (Low-Rank Adaptation) [213].

Incremental learning: To make sure that LLMs keep up with an evolving knowledge base, it is often not enough to train them once, but update them over time. Jovanovic et al. [137] considered different strategies for an incremental learning of LLMs. These include continual learning (CL), meta-learning, parameter-efficient learning, and mixture-of-experts learning.

Shi et al. [272] conducted a comprehensive survey on CL. Here, approaches are divided in two categories: vertical and horizontal continuity. Vertical continuity addresses approaches that specialize capabilities from a general set of knowledge. Horizontal continuity describes approaches that adapt capabilities across time and domains. In addition to outlining CL approaches, they included background information on CL, training objectives, as well as an overview of benchmarks.

Wu et al. [341] showed that CL can be used to update several dimensions: facts, domains, language, tasks, skills, values, preferences. Yang et al. [374] took pre-trained, fine-tuned, and vision-language models in account and CL methods are split into offline and online methods. In addition to internal methods for CL, such as the updating of parameters, Zheng et al. [419] included external approaches in their survey. External knowledge can either be incorporated by retrieving information from websites (e.g., Wikipedia), or the use of tools to allow LLMs to carry out additional tasks.

Unlearning: Learning can help LLMs attain valuable capabilities but not all the information might be useful to learn. Among others, LLMs might learn biases or access private information of individuals in the training data, which should not be replicated. Unlearning approaches are proposed to help LLMs forget about undesired information. The survey by Blanco-Justicia et al. [16] presented different types of unlearning approaches with regard to global weight modification, local weight, architecture modification, and input or output modification. They also showed datasets, models, and metrics used for evaluation. Xu [360] considered unlearning traditional ML models and LLMs, while Qu [251] surveyed unlearning approaches for federated learning.

4.5 Alignment

Via pre-training and fine-tuning, LLMs are capable of learning from data and generating sensible responses for a variety of tasks. However, such responses can be factually incorrect or harmful due to undesired biases in the training data [84, 326]. To combat this, alignment approaches are proposed not only to align LLM responses with human values but also restrict their misuse in sensitive or potentially harmful contexts.

Wang et al. [326] focused on alignment techniques, such as reinforcement learning from human feedback. They surveyed different stages of the reinforcement learning process and included equations to explain the respective techniques. Shen et al. [271] divided alignment approaches into

“outer” and “inner” approaches. Outer alignment describes the alignment of LLMs to human values. Inner alignment concerns the optimization of the objectives humans desire. Cao et al. [22] reviewed automated alignment methods, which do not require human annotations. Four method types are presented: aligning by defining constraints, imitating behavior of an aligned model, receiving feedback by existing models, and through feedback from interacting with environments.

Other than the alignment approaches, surveys considered their evaluation [197] and the collection of high-quality datasets [325]. Moreover, alignment was studied for agents [98] and the 17 United Nations’ Sustainable Development Goals [340]. Other than the alignment for a general audience, the alignment of LLMs to individuals has been surveyed from the perspective of personalization [147] and preference learning [128].

In addition to aligning LLMs to human values, their adaptation of roles and pre-defined personas has been studied. Particularly, Tseng et al. [296] surveyed LLMs for role-playing and personalization. Here, LLM role-playing was performed for tasks such as software development, games, and the medical domain. The survey by Chen et al. [32] outlined how to design components for role-playing, which includes datasets and different alignment methods. Simmons and Hare [278] gave an overview of LLMs to replicate human behavior, such that public opinions can be measured.

4.6 Prompting

LLMs are able to respond to user queries, therefore, the design of such queries or prompts can be integral for achieving good results. Chen et al. [29] created a review on prompt engineering, which could be seen as a good starting point for someone trying to create prompts themselves, as they provided basics on prompting models such as GPT-4. Luo et al. [205] considered prompting from the perspective of in-context learning, which can achieve performance improvement by providing LLMs with examples. Important aspects include the number of demonstrations as well as their diversity and order [427]. The surveys by Vatsal and Dubey [302] and Sahoo et al. [264] set out to provide an overview of diverse sets of prompts used in the literature. In total, Vatsal and Dubey [302] presented 39 different prompting methods, Sahoo et al. [264] showed 29.

Focusing only on a subset of approaches, Xia et al. [346] created a survey on chain-of-thought prompts and its variations (chain-of-X). Approaches applied in literature can be divided into four groups: chain-of-intermediate, chain-of-augmentation, chain-of-feedback, chain-of-models. Other than chain-of-thought approaches, five surveys focused on Retrieval-augmented generation (RAG) prompt approaches [66, 80, 112, 120, 136]. Huang and Huang [120] created a taxonomy of RAG approaches based on four stages. First, pre-retrieval approaches deal with the data, followed by the retrieval stage which ranks the relevant pieces of information. Subsequently, a post-retrieval stage can re-rank and filter the retrieved information. In the end, the generation stage creates the prompt. Jing et al. [136] carried out a survey with a focus on the pre-retrieval stage, in particular the organization of data in vector databases to support the subsequent retrieval stages. The survey by Gao et al. [80] considered the latter three stages of the RAG process (retrieval, generation and augmentation). Their survey included RAG approaches at different stages of LLMs (pre-training, fine-tuning, inference) and progression of RAG from naive to complex approaches. Moreover, they extracted useful information from the approaches, such as the data sources considered, data type and retrieval granularity (e.g., sentences). Hu and Lu [112] pointed out approaches which combined multiple RAG techniques (i.e., hybrid approaches). Additionally, surveys also cover prompting for specific domains, such as reasoning [243], software engineering tasks [17], visual-language models [90], and goal-oriented tasks [166].

While prompt engineering can lead to performance improvements, it also requires additional efforts, either from the developers to design the prompts or from additional computations carried out by the LLMs. For this purpose, Chang et al. [25] surveyed the efficiency of prompt engineering from

two perspectives: prompting with efficient computation and prompting with efficient design. Lastly, Liu et al. [193] surveyed prompting frameworks, or prompt-based tools, which can simplify the prompt engineering process for users. Their overview includes and compares 28 such frameworks.

4.7 Agents

An LLM-based agent employs an LLM as the primary component of the controller module in an attempt to achieve goals autonomously [343]. Several general surveys aim to establish a common taxonomy for this field. Some researchers converge on a core set of components such as reasoning, memory, and action modules for tool use and interaction with the environment [13, 414]. Others, such as Wang et al. [313] proposed four modules: profiling, memory, planning, and action, while Xi et al. [343] conceptualized agents through a simplified brain-perception-action framework. From a different perspective, Li [171] proposed to integrate the previously separated paradigms of tool-using, planning, and feedback learning, into a task-agnostic taxonomy through three universal roles and four workflows.

Building upon these general frameworks, some researchers survey specific modules or capabilities. Huang et al. [118] surveyed and proposed a taxonomy on the planning module classifying them into five directions. Similarly, Zhang et al. [409] reviewed memory mechanisms and how they enable self-improvement and long-term interactions. The capabilities further extend when these systems encompass multiple agents. Guo et al. [92] reviewed LLM-based multi-agent systems, highlighting communication and planning strategies in different environments, while Händler [98] proposed a taxonomy for classifying these systems based on alignment, autonomy, and architectural design.

The applications of LLM-based agents are rapidly expanding through different tasks and fields. Hu et al. [111] proposed a general taxonomy for agents into six modules and further reviewed their applications in the gaming field. Cao et al. [23] explored how LLM-based agents can be used to enhance reinforcement and classified them according to their function in the framework. Gao et al. [77] focused on their application on simulation and modeling of scenarios in four domains: cyber, physical, social, and hybrid. Social simulations are also reviewed by Ma et al. [207], who focused on computational experiments to generate more realistic social behaviours. Ramos et al. [255] explored applications in chemistry, discussing automation of tasks such as literature review to controlling robotic labs. Li et al. [177] reviewed the surge of personal LLM-based agents and discussed the security threats when handling personal data. Similarly, the risks and threats introduced by the integration of agents in multiple fields are further explored by He et al. [101].

4.8 Inference

The inference process has in general been covered by surveys under the lens of efficiency. Addressed techniques can be divided into two categories: model compression and dynamic acceleration. They are summarized in Table 3.

Model compression aims for efficiency improvements by lowering the number of parameters and thereby achieving quicker responses. Park et al. [232] surveyed compression methods, which include four common methods (pruning, quantization, knowledge distillation, low-rank approximation), and they are one of two studies which considered parameter sharing. They distinguished compression methods based on their cost (i.e., low-cost and high-cost) depending on how much time and memory their use requires. For example, high-cost quantization methods might require a full retraining of an LLM while low-cost quantization can be performed without any retraining or fine-tuning. Lastly, they provided additional evidence on the usability of pruning and quantization methods by evaluating them and summarizing their cost, compression rate and performance. Chavan et al. [28] included empirical results on several quantization types and their impact on memory consumption and number of tokens that get generated per second.

Table 3. Overview of inference techniques and surveys which describe them.

Model Compression	Pruning [9, 28, 55, 134, 232, 260, 305, 320, 351, 356, 386, 428, 430]
	Quantization [9, 28, 134, 232, 260, 305, 320, 351, 356, 386, 428, 430]
	Knowledge distillation [9, 28, 55, 134, 232, 305, 320, 351, 356, 365, 386, 428, 430]
	Low rank approximation [9, 28, 232, 305, 351, 356, 386, 428, 430]
	Parameter sharing [232, 351]
Dynamic acceleration	Early exit [9, 145, 351, 386, 428]
	Input pruning, filtering and compression [9, 356, 428]
	Token parallelism [9, 145]
	Token skipping [351]
	Speculative decoding [145, 305, 344, 386, 391, 428]
	Output organization [428]
	MoE [320, 386]

Commonly, surveys considered at least three of the five compression methods, while we encountered two surveys that covered a single method. In both cases, this was Knowledge Distillation (KD) [359, 365]. Xu et al. [359] provided extensive information on the KD process to help understanding. Moreover, they described how KD can be used to distill skills (e.g., instruction following, alignment) and domain knowledge (e.g. law, science). Yang et al. [365] categorised approaches as white-box and black-box, and also presented KD approaches for multi-modal LLMs. Additionally, they included experiments with KD methods on different LLMs and compared their performance.

Dynamic acceleration refers to techniques that adapt computations based on the input the LLMs receive [9]. While it is common for surveys to not be focused on any of the dynamic acceleration techniques in detail, we found two surveys on speculative decoding, a method that predicts several future tokens alongside the current token at each decoding step [344, 391].

4.9 Evaluation

Evaluating the performance of LLMs is a critical step in assessing their quality. Chang et al. [27] investigated existing evaluation approaches based on three perspectives: what to evaluate (e.g., tasks), where to evaluate (e.g., datasets, benchmarks), how to evaluate (e.g., evaluation process). Here, tasks include NLP, social science, natural science, engineering, medical applications, robustness, and others. Benchmarks are divided into three categories: general, specific, and multi-modal. Lastly, the evaluation process can follow automatic or human evaluation criteria.

Guo et al. [94] defined three categories of LLM evaluation: knowledge and capability (e.g., reasoning), alignment evaluation (e.g., bias), and safety (e.g., robustness). They reviewed each category and provided benchmarks and evaluation methodologies. Moreover, they present LLMs specialized for various domains (e.g., biology, medicine, finance, education, computer science).

Zhuang et al. [432] structured the evaluation of LLMs based on four core competencies (i.e., knowledge, reasoning, reliability, and safety) and provided their definitions, benchmarks, and metrics. In addition to core competencies, it is important to consider sentiment, planning, and code. Beyond the evaluation of competencies, Peng et al. [235] considered the evaluation of agentic approaches (e.g., reasoning, domain knowledge).

A different focus was set by Laskar et al. [159], who put emphasis on challenges and limitations of the evaluation caused by the diversity of setups followed by existing works. In particular, they considered three types of challenges: reproducibility (e.g., missing details on data and model), reliability (e.g., prompt hacking), and robustness (e.g., lacking evaluation of generalizability). Additionally, Laskar et al. [159] gave guidelines on how these limitations can be addressed.

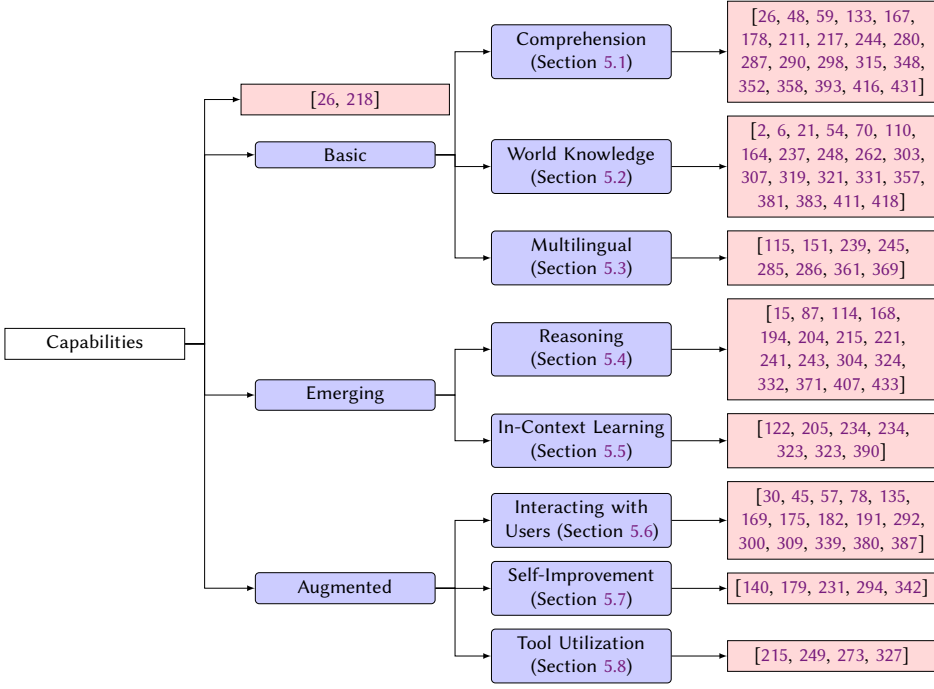


Fig. 5. Taxonomy of surveys on capabilities.

5 Capabilities

The proficiency of LLMs in diverse tasks and fields can be understood as a result of multiple abilities, or capabilities. These capabilities range from foundational language skills to more complex behaviors. While Chang and Bergen [26] surveyed capabilities of language models, this section adopts a taxonomy inspired by the work of Minaee et al. [218], as it further covers how models are integrated into larger ecosystems. This framework categorizes these abilities into three groups: Basic, Emerging, and Augmented.

Basic capabilities represent foundational skills acquired directly from their pre-training. Emerging capabilities are not specifically trained for, but appear as the scale of the model increases. Augmented capabilities regard how the model integrates with external systems, usually to overcome its own limitations and improve the outcomes.

5.1 Comprehension

The comprehension capability of LLMs refers to their ability to understand and interpret the meaning, context, and possibly the nuances of input. It is crucial for their application to various NLP tasks [26, 217, 244], such as information retrieval [416, 431], relation [358], keyphrase extraction [280], semantic parsing [211], question-answering [48], and text generation [167, 178, 393].

Several surveys focus on more specific types of LLM comprehension. For instance, the models' ability to understand the content is essential to produce accurate summaries. General text summarization surveys [133, 290] indicate that abstractive summarization requires the model to interpret and rephrase information, which allows for testing the depth of the comprehension. Xie et al. [348] further corroborated this in the domain of biomedical texts. Other authors use different tasks to

test this comprehension. Umair et al. [298] surveyed this capacity with keyphrase prediction, Xu et al. [352] focused on information extraction, and Sun et al. [287] used sentiment analysis.

The underlying mechanisms enabling LLMs to comprehend texts based on previous knowledge are also an ongoing research field. Wang et al. [315] discussed how true comprehension allows for high-level reasoning and problem-solving skills to develop. They further pointed out the fragility of comprehension in LLMs, agreeing with the main challenge presented by Du et al. [59].

However, the depth of this comprehension remains ongoing research. Du et al. [59] indicated that LLMs can perform “shortcut learning”, where they overfit to dataset biases rather than achieve proper understanding of the text, leading to poor performance in out-of-distribution scenarios.

5.2 World Knowledge

We refer to world knowledge capability in LLMs as their ability to store and accurately recall factual information acquired during their pre-training phase. This capability is fundamental to the use of LLMs, but also involves challenges regarding its representation, robustness, and maintenance.

The current understanding indicates that models learn universal language representations that capture knowledge about the world during their pre-training phase and are stored implicitly in the model’s parameters [248, 381, 411] and can be used as knowledge bases [6]. Cao et al. [21] conceptualized the knowledge in an LLM through a life cycle consisting of five phases: acquisition, representation, probing, editing, and application. On the other hand, Safavi and Koutra [262] conceptualized this knowledge through different levels of Knowledge Base (KB) supervision.

The accuracy of the knowledge assimilated by the LLM is still an ongoing concern. Adilazuarda et al. [2] noted that knowledge assimilated by LLMs often exhibits biases. Wang et al. [307] focused on a different issue, factuality, i.e., generating content contradicting facts. Zhang et al. [411] argued that since facts learned during pre-training can become outdated, updating this knowledge also poses a problem. Similarly, the meaning of words can change over time, as shown in the survey by Periti and Montanelli [237]. Updating the models with contextual information is a possible solution; however, Xu et al. [357] pointed out that models also struggle with discrepancies between learned facts and contextual information.

Researchers are exploring different techniques to address these limitations by integrating the models with desired knowledge [70, 110], which can come from different sources and formats [418]. Wang et al. [319] reviewed knowledge editing techniques, aiming to update knowledge without complete retraining. On the other hand, Yin et al. [381] and Wei et al. [331] advocated for augmenting LLMs with external sources of knowledge. Wang et al. [321] further explored this synergy with a survey of LLMs enhanced with knowledge representation learning.

In addition to teaching facts to LLMs, probing can be used to extract facts [383], and LLMs have been studied for fact-checking [54, 303]. To support the ability of fact-checking, the generated texts can be attributed with respective sources [164].

5.3 Multilingual

The multilingual capabilities of LLMs enable them to understand and generate text in multiple languages. This ability facilitates knowledge transfer from high-resource to low-resource languages [115, 151, 361]. Generally, multilingualism in a model is achieved through multilingual corpora during pre-training [239, 361]. Xu et al. [361] surveyed multilingual LLMs and gathered valuable resources for corpora (training and downstream tasks) as well as LLMs. For the datasets, they showed the languages in the corpora, the sizes, and sources. For LLMs, they extracted information such as the base model, number of parameters and pre-training information (dataset and loss function). Moreover, they carried out a performance comparison of several models over three tasks (bilingual lexicon induction, cross-lingual classification, machine translation) and ten languages.

Qin et al. [245] focused on data resources and alignment techniques needed to achieve this capability, distinguishing between parameter-tuning and parameter-frozen approaches. The survey by Huang et al. [115] extended beyond the contents of Qin et al. [245] and Xu et al. [361]. While Qin et al. [245] provided a comprehensive taxonomy for alignment techniques, Huang et al. [115] devised a taxonomy spanning training, inference, security, domains (medical, legal), and datasets. Similar to Xu et al. [361], Huang et al. [115] provided comprehensive overviews for datasets (training and benchmarking) and available models. In contrast to Xu et al. [361], only models with 7B and more parameters have been listed. Lastly, Huang et al. [115] provided insights on security via attack (e.g., jailbreaking) and defense methods.

In contrast to the broad scope of multiple languages, other surveys have focused on a single language. For high-resource languages such as Chinese, Sun et al. [286] provided an extensive review of LLMs in different downstream applications such as sentiment analysis. Similarly, Subies et al. [285] focused on surveying clinical LLMs for Spanish. On the other hand, surveys on lower resource languages tend to focus on establishing a benchmark for foundational capabilities. Yang [369] surveyed and benchmarked the performance of available LLMs in Korean. Addressing the challenges of languages with even fewer resources, Kryeziu and Shehu [151] reviewed techniques to build LLMs for low-resource languages, using Albanian as a key example.

While the previous surveys have focused on how this capability is acquired, Philipppy et al. [239] underscored the lack of explanation for this capability and performed a review on possible factors. In total, they considered five factors: linguistic similarity, lexical overlap, model architecture, pre-training settings, and pre-training data.

5.4 Reasoning

We refer to reasoning as the capability to perform deductions, make decisions, and conduct multi-step problem solving. This capability is crucial for LLMs to move from simple pattern matching and produce more refined inferences able to tackle complex tasks.

This reasoning has significantly increased through prompting techniques such as Chain-of-Thought (CoT) [215, 241, 243]. Li et al. [168] discussed general fundamental capabilities of LLMs, further decomposing the reasoning step. Huang and Chang [114] surveyed a general method to elicit and evaluate reasoning abilities, while Qiao et al. [243] and Laat et al. [241] proposed taxonomies for prompt-based reasoning.

Similarly to previously discussed capabilities, evaluating reasoning still poses a challenge [221]. Some surveys have researched specific branches of this capability; Luo et al. [204] focused on logical reasoning, Wang et al. [324] researched multimodal reasoning, and Zhang et al. [407] studied strategic reasoning. Some authors have evaluated reasoning through tasks: Giadikiaroglou et al. [87] explored evaluations through puzzles, Zong and Lin [433] used categorical syllogisms, Wen et al. [332] studied answer refusals, and Bhargava and Ng [15] involved commonsense knowledge. However, Yang et al. [371] argued that these benchmarks may be solved through knowledge retrieval. Lastly, Liu et al. [194] surveyed the applicability of LLMs and their reasoning capability to aid with causal inference, or causal discovery [304].

5.5 In-context Learning

In-Context Learning (ICL) refers to the ability to adapt to new tasks by processing examples in the LLM input. This allows for quick, few-shot adaptation to new tasks without modifying the model's parameters. Luo et al. [205] provided a general survey focusing on retrieval-based ICL.

The limited context window of LLMs poses a challenge for ICL, by limiting the number of examples that can be used in the prompt [234, 323]. Huang et al. [122] surveyed progress in LLMs aimed at increasing their context. Similarly, Pawar et al. [234] and Wang et al. [323] surveyed

techniques for extending context length. However, Zeng et al. [390] underscored that context length is only one of the three conflicting goals, the other two being accuracy and performance.

5.6 Interacting with Users

LLMs are increasingly deployed in applications that require direct user interaction. This interaction requires abilities to engage in dynamic conversation, understand the intentions from the user, and generate helpful responses [309, 380]. Zaib et al. [387] performed a general survey on dialogue systems with LLMs, exploring how they can be leveraged for conversational agents. Similarly, Dam et al. [45] analyzed LLM-based chatbots and their impact on diverse fields. Gao et al. [78] devised four stages for the interaction between humans and LLMs. They consist of: planning, facilitating, iterating, and testing. Focusing on the progression of these systems, Wang et al. [309] provided a deeper assessment on the evolution and trends of LLM-based dialogue systems. One key aspect in this evolution, multi-turn dialogues, was surveyed by Yi et al. [380]. However, Dong et al. [57] stated that as these models become more integrated into user-facing applications, ensuring their safety and robustness becomes essential.

Beyond the focus on dialogue, LLMs are used to understand user needs and characteristics. This area of research is known as user modeling. Tan and Jiang [292] described how LLMs are used to model and understand user-generated content. Jin et al. [135] surveyed how LLMs can infer the user background based on cues in the prompts and tailor their responses accordingly. Another avenue of interaction between LLMs and users is recommendation systems. In this context, the survey by Li et al. [175] provided background details on ML and DL-based recommendation, comparing them to LLM-based approaches. Lin et al. [182] and Vats et al. [300] presented how and what parts of a recommendation system can be supported by LLMs. Wu et al. [339] categorized LLM approaches for recommendation in two paradigms: discriminative (generating embeddings for users and items) and generative. Other than computing scores for items, generative recommendation directly generates recommendations. This can be achieved by representing user and item IDs via tokens. Generative recommendation was examined in more detail by two more surveys [169, 175].

Other surveys have focused on practical aspects of building these LLM-based recommender systems. For instance, Liu et al. [191] examined the training strategies for LLMs in recommendation tasks, describing learning objectives, data types, and datasets used in each publication. Lastly, Chen [30] surveyed how to generate explanations for recommendations and accompanying challenges.

5.7 Self-Improvement

Self-Improvement encompasses the capability of LLMs to learn from feedback to autonomously enhance their results. Two surveys offer a broad view: Pan et al. [231] classified self-correction strategies according to when the correction occurs (training, generation, post-hoc). Tao et al. [294] used the concept of “self-evolution” and broke it down into a four-phase iterative cycle: experience acquisition, experience refinement, updating, and evaluation.

Other surveys go into a deeper analysis of the self-improvement process. Kamoi et al. [140] claimed self-correction results are being overstated due to unfair evaluation. They concluded that reliable feedback is often the bottleneck, indicating that self-correction without external tools generally fails except for suitable tasks. The unreliability of self-feedback is further discussed by Liang et al. [179], who connected the success of self-improvement to internal consistency. They concluded that since LLMs are trained on mostly correct data, improving the consistency of their outputs tends to increase the probability of a correct output more than an incorrect one. Lastly, Wu et al. [342] considered how evolutionary algorithms can be used to enhance LLMs by supporting them with search capabilities, while LLMs can be used to enhance evolutionary algorithms by guiding the search process with domain knowledge.

5.8 Tool Utilization

LLMs are capable of using tools with the goal of interacting and leveraging external programs, such as external software and APIs, to overcome limitations and perform additional functionalities. This capability allows the LLMs to solve more complex problems and interact with the environment.

Wang et al. [327] performed a general survey and proposed a taxonomy for tools based on their functionality. Qu et al. [249] surveyed tool utilization and proposed a four-stage workflow for tool learning. Other authors survey specific applications in this field. For example, Shi et al. [273] examined the use of tools after the content is generated by focusing on Text-to-SQL tasks, and Mialon et al. [215] studied the use of other models, search engines, and the web as tools.

6 Applications

LLMs have shown promise in various applications and industries [299], ranging from critical fields (e.g., finance, health, law) [35], to niche topics such as fitness or climate modeling [142]. This section outlines the main application domains in which LLMs have been used, and their respective surveys.

6.1 Medical and Health

Medical and health applications are the most popular domain for LLM surveys we encountered, with a total of 32 surveys carried out up to September'24.

Xiao et al. [347] and Zhou et al. [422] created surveys containing information about the training, data and applications for LLMs in the medical domain, as well as challenges and areas for future research. Both surveys provided helpful overviews of datasets and models, with information such as the base model and data source, where Xiao et al. [347] also took multimodal LLMs into account. In total, Xiao et al. [347] considered six applications: medical diagnosis, clinical report generation, medical education, mental health services, medical language translation, and surgical assistance. The set of applications studied by Zhou et al. [422] shows some overlap; however, the fields of medical robotics, clinical coding, medical inquiry, and response are novel.

Similarly, Wang et al. [306] considered vision and standard LLMs for pre-trained models and fine-tuning for downstream tasks. Luo et al. [206] focused their survey on pre-trained LLMs for NLP tasks. Their overview included English and Chinese LLMs used for various tasks, such as question-answering, machine translation, sentiment analysis, and named entity recognition. For each task, they provided details on datasets and metrics used.

He et al. [102] transitioned from PLMs to LLMs. This included details on training and datasets. Similarly, Wang et al. [311] covered the data acquisition process and different training paradigms to adapt general LLMs for the medical domain. Their survey also included concerns about fairness, accountability, transparency and ethics. Park et al. [233] considered ethical implications in their review, as well as legal and socioeconomic concerns. In addition to a comprehensive overview, Liu et al. [190] put emphasis on trustworthiness and safety of LLMs, which includes a discussion of their fairness, accountability, privacy, and robustness. Several other surveys considered privacy and ethical concerns in the medical domain [96, 224, 252, 420].

Huang et al. [121] focused on the evaluation of medical LLMs. This included evaluation approaches and metrics for different applications: departments and specific diseases, medical research, medical education and public awareness, and medical text processing. LLMs in the medical domain have been evaluated by three different evaluators: human experts, automated metrics, and AI-driven assessments. Automated metrics can be categorized in four groups: correctness, completeness, usability, and consistency. AI-driven assessments are in the minority. Chen et al. [33] also considered the evaluation of LLMs for medical tasks such as image processing and information extraction.

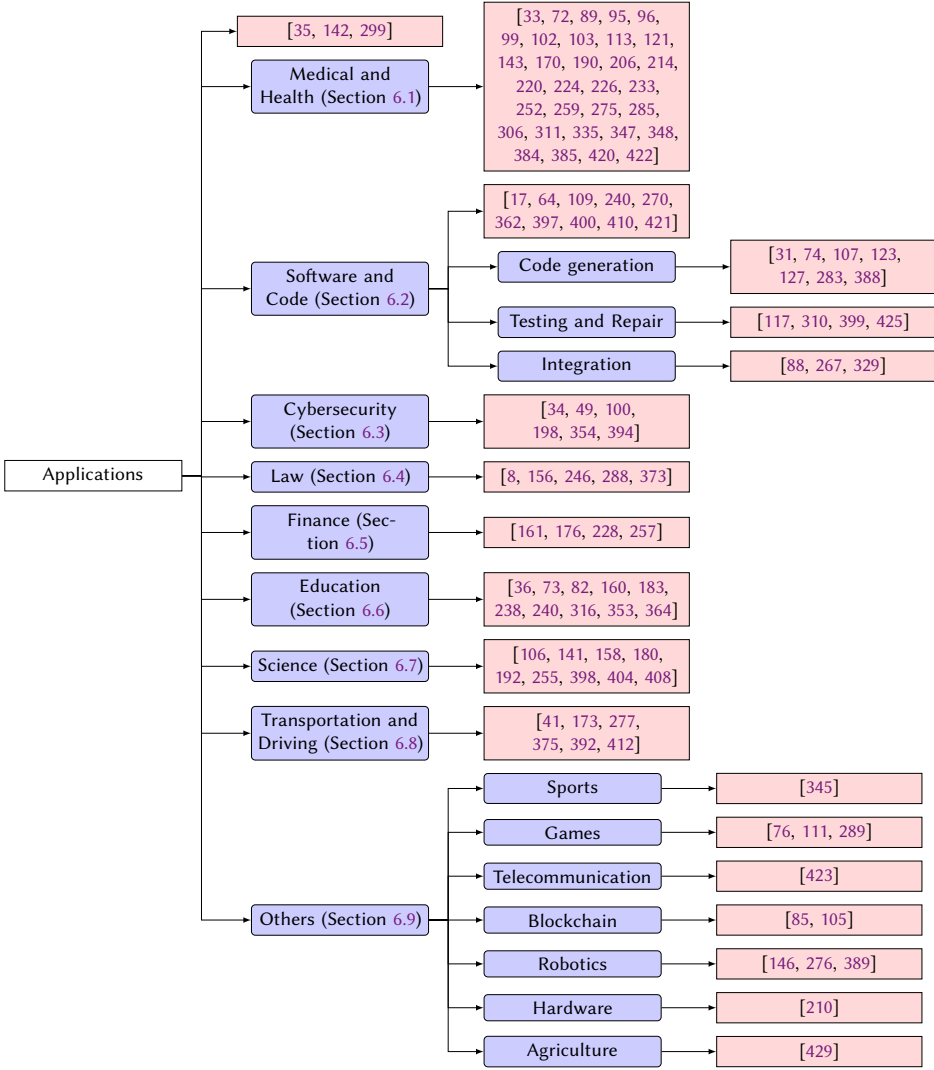


Fig. 6. Taxonomy of surveys on applications.

While a lot of the surveys described tasks based on texts, the field of medicine is multimodal and several types of data have been surveyed [99, 306, 347, 385]. Ferrara [72] studied data collected by wearable sensors and the survey by Nerella et al. [226] covered data types such as NLP, medical imaging, structured Electronic Health Records (EHR), social media, biophysiological signals, and biomolecular sequences. Particularly, electronic health records have been of interest for surveys [170, 335, 348]. Li et al. [170] surveyed LLMs working with Electronic Health Records, in particular with regards to seven tasks: named entity recognition, information extraction, text summarization, text similarity, text classification, dialogue system, diagnosis, and prediction. Xie et al. [348] only considered the task of text summarization, which has been applied for EHR and biomedical literature, medical conversation, and questions.

Another set of surveys included bibliometric analysis. For instance, Restrepo et al. [259] analyzed metadata such as author affiliations, countries, and funding source to assess diversity. Yu et al. [384]

considered information such as collaboration networks. The remaining surveys covered areas ranging from only considering Spanish language models [285] to LLMs in medical examinations[220], psychology [103, 143], mental health [89, 95, 113, 214], and critical care medicine [275].

6.2 Software and Code

In the software engineering domain, we found several surveys which provided comprehensive overviews. The earliest survey is by Xu and Zhu [362], from 2022. They surveyed datasets, tasks, and architectures for pre-trained LLMs as well as their training procedures.

Subsequent surveys increased in comprehensiveness, with the survey by Ziyin Zhang et al. [410] covering more than 900 works. They created both a taxonomy for code LLMs as well as a taxonomy for more than 40 tasks according to the software development stages. The survey by Qianjun Zhang et al. [400], which also entails more than 900 references, provided another comprehensive overview. Interesting aspects they considered included an overview of pre-training tasks as well as the integration of LLMs for SE activities (e.g., their security or size).

Zheng et al. [421] gave information about organizations which developed the LLMs (e.g., Company-led, University-led, Research teams & Open-source community-led). Also, their survey put emphasis on the performance of LLMs. One research question was aimed at finding whether code LLMs perform better than general LLMs for SE tasks. Moreover, they presented the performance reported in collected works for several tasks, to find which LLM is most suitable. Hou et al. [109] provided valuable insights on the datasets used for SE tasks, including data collection, selection, and processing steps. She et al. [270] surveyed pitfalls which could hinder the performance of LLMs in practice. These are divided into five categories: data collection and labeling, system design and learning, performance evaluation, deployment, and maintenance. For each of these pitfalls, implications and solutions are outlined. Similarly, Fan et al. [64] listed open problems for each stage of the software development lifecycle.

Other surveys investigated how LLMs have been prompted for various SE tasks [17], how LLMs can be used in an educational setting to help with code related tasks (e.g., explaining error messages) [240], or support failure management for Artificial Intelligence for IT Operations [397]. **Code generation:** Jiang et al. [127] created a comprehensive survey on the generation of code from natural language descriptions. Collected works are structured given a taxonomy in: data curation, recent advances (e.g., training and prompting), evaluation, and application (e.g., GitHub Copilot). They also provided an overview of existing LLMs and a performance comparison of several LLMs on two popular benchmarking datasets: HumanEval and MBPP. Zan et al. [388] also provided a comparison of LLMs on the HumanEval benchmark, where they included a larger quantity of small LLMs (smaller than 1 billion parameters). Additionally, they presented 17 benchmarks with statistics, such as the number of tests available. In contrast, Hong et al. [107] surveyed approaches for generating SQL queries from natural language.

Husein et al. [123] surveyed the completion of code rather than generating code from natural language descriptions. They considered different granularities (token, line, API calls, Block level) and performance metrics for evaluation. Other than generating code itself, LLMs have been used to generate programming exercises [74], infrastructure configurations [283], and support HPC [31]. **Testing and Repair:** The survey by Wang et al. [310] discussed the field of software testing and different associated tasks. The most commonly addressed tasks include program repair as well as the generation of tests (e.g., unit tests, system tests). For these, Wang et al. extracted the most common prompts (e.g., zero-shot) and the LLMs used for these tasks.

The survey by Zhang et al. [399] focused on APR and found 127 APR papers covering 18 bug types that used LLMs. Zhou et al. [425] considered both vulnerability detection and repair. They

investigated how LLMs have been adapted to these tasks and found that the majority of approaches perform fine-tuning. Huang et al. [117] covered the use of LLMs for fuzzing as a testing activity.

Integration: While previously outlined surveys covered the use of LLMs for software engineering activities, they can also be treated as components of software itself. In this regard, Weber [329] created a taxonomy for LLM-integrated software systems, and Sergeyuk et al. [267] studied the use of LLMs in Integrated Development Environments (IDEs). Gorissen et al. [88] considered the use of LLMs in Low-Code Development Platforms.

6.3 Cybersecurity

Four studies created comprehensive overviews of the field of cybersecurity [34, 100, 354, 394]. Hasanin and Moustafa [100] covered diverse cyber defense strategies such as vulnerability assessment, intrusion detection, or anonymization, while others put emphasis on vulnerability assessments [49] and threat detection [34]. Zhang et al. [394] not only outlined defense activities but also indicated ways to use LLMs for attacks. Xu et al. [354] provided insights on how to construct LLMs for the security domain via means of fine-tuning, prompting, or augmentation with external tools. Lastly, Liu [198] gave an overview of available pre-trained models for cybersecurity, and Xu et al. [354] addressed the data collection process and available datasets.

6.4 Law

LLMs have been used to automate various legal tasks, but their adoption also raised challenges [288]. Anh et al. [8] researched the impact of LLMs on NLP, focusing on legal text processing. They explained how NLP addresses different challenges in the field, such as ambiguity and sentence complexity. They performed an empirical analysis that suggests that encoder-decoder models outperform encoder-only architectures, advocating for their use in legal NLP tasks.

Lai et al. [156] provided a general survey on the applications of LLMs within the judicial systems. They included the impact on common users as well as experts (e.g., judges and lawyers). The authors indicated limitations and issues of LLMs that can affect judicial practices. They gave practical recommendations for improving the use of LLMs in the legal system and highlighted the importance of understanding the societal impacts of these technologies. Similarly, Qin and Sun [246] covered the practical application of LLMs in the legal system, such as case retrieval and legal analysis. They indicated potential challenges such as biases, interpretability issues, and data privacy concerns. This study emphasized the need for fine-tuned models and presented an overview of datasets for their training in different languages. Lastly, Yang et al. [373] presented a systematic review of legal LLMs focusing on fine-tuning for question-answering tasks. They provided a practical view focusing on the implementation of these systems and the techniques that they could use (e.g., Low-Rank Adaptation). They used a bottom-up approach to examine how existing models can be adapted to the legal domain.

6.5 Finance

Nie et al. [228] provided a comprehensive survey on LLMs for finance. They first categorized existing works according to application areas in the financial domain, including, among others, time series forecasting, reasoning, and sentiment analysis. Further information on datasets, benchmarks, and challenges is presented. In addition to providing an overview of finance applications, Li et al. [176] developed a decision framework to help practitioners select an LLM based on their task. For this, they also provided a comparison with estimated costs of different LLM options (e.g., zero-shot, fine-tuning, training from scratch). Lee et al. [161] put emphasis on presenting benchmark tasks and datasets. Moreover, they showed a timeline of LLMs and financial LLMs. Ren et al. [257] addressed

the use of LLMs in an e-commerce setting. In this context, LLMs have been used for tasks such as product recommendations, question answering and analysis of customer feedback.

6.6 Education

Wang et al. [316] created a comprehensive survey on how LLMs can assist teachers, students, and different tools that are available. Additionally, they provided an overview of datasets and benchmarks, as well as discussed risks and challenges of LLMs in education. Pester et al. [238] addressed the use of LLMs for immersive learning activities. The survey by Xu et al. [353] provided more background information on education, as well as how to integrate LLMs in the process, while García-Méndez et al. [82] considered LLMs used for different education activities. This focus on integration also extends to specific disciplines, with dedicated surveys exploring the use of LLMs in subjects such as computer science [240] or engineering [73].

Yan et al. [364] covered a total of 53 educational tasks from nine categories (e.g., grading, content generation) and put emphasis on practical and ethical challenges. In a similar fashion, Chhina et al. [36] looked at both the challenges and benefits of LLMs in education. Lee et al. [160] focused their survey on different types of biases when using LLMs in an educational setting. Biases were investigated at different stages of the LLM lifecycle (e.g., data collection, training, and deployment). Lin et al. [183] listed available open-source LLMs for use in education activities.

6.7 Science

Ho et al. [106] provided an overview of scientific LLMs applied to text, and presented different tasks, datasets, and existing models. In addition to scientific LLMs for text, Zhang et al. [404] surveyed more than 260 LLMs, not only taking different scientific fields but also different modalities into account. Complementing this broad overview, other surveys focus on LLM applications in specific fields such as chemistry [180, 255, 398], biology [398], and mathematics [192], as well as for specialized sub-domains like single-cell biology [158] and computational neuroscience [141].

A trait of scientific texts is the presence or use of citations, to give credit to relevant sources. Here, Zhang et al. [408] created a survey to show the relation between LLMs and citations. Their survey provided an overview of four different citation tasks LLMs can be applied to: citation classification, citation-based summarization, citation sentence generation, and citation recommendation. Additionally, they discussed how citations can be incorporated in the training of LLMs.

6.8 Transportation and Driving

In the realm of Intelligent Transportation Systems (ITS), LLMs have been used to advance transportation intelligence and traffic management. The surveys by Shoaib et al. [277] covered tasks such as traffic prediction and transportation management, while Zhang et al. [392] considered traffic management, transportation safety, and autonomous driving. Moreover, they provided a list of datasets for the ITS domain. Zhang et al. [412] focused on travel behavior prediction as a time series forecasting problem and provided an overview of LLM-based approaches.

Autonomous driving was covered by three dedicated surveys [41, 173, 375]. Cui et al. [41] addressed the use of LLMs for autonomous driving from a multimodal perspective (vision and language). They provided a holistic overview, considering the use of multimodal LLMs for autonomous driving, transportation, and maps. Furthermore, they presented datasets for autonomous driving and traffic scene understanding, and extracted information from existing approaches, such as the LLMs used. In contrast, Yang et al. [375] provided a more fine-grained view on tasks and metrics used for evaluation. They distinguish four categories, based on the respective tasks: planning, perception, question answering, and generation. Li et al. [173] covered the use of LLMs in autonomous driving either as part of the pipeline, to support existing systems, or as end-to-end systems.

6.9 Others

Sports: Xia et al. [345] investigated datasets and applications for LLMs in a sports setting. Here, LLMs can be applied to different input types (text, video, audio) and have addressed a diverse range of tasks (e.g., hate speech detection, fan engagement, game summarization).

Games: Sweetser [289] carried out a scoping review on 76 papers on LLMs for video games, to provide an overview and support future research. In the gameplaying context, LLMs have been used as parts of the game (e.g., agents, dialogue generation) or part of the development and analysis process (e.g., content generation, analysis of reviews).

Gallotta et al. [76] addressed the different roles of LLMs in games in their survey. In total, they identified nine roles that LLMs have taken: Player, NPC (non-player character), player assistant, commentator, analyst, game master, game mechanic, automated designer, and design assistant. Additionally, they presented a roadmap for future applications of LLMs for games, as well as limitations and ethical implications of their use.

Hu et al. [111] focused their survey on LLM-based game agents, for which they found 62 approaches. These were categorized based on game type (text, video) and genre (e.g., adventure, cooperation, simulation). Moreover, agents were discussed from six perspectives: perception, memory, thinking, role-playing, action, and learning.

Telecommunication: Zhou et al. [423] presented a comprehensive overview of LLMs in the field of telecommunications. In particular, LLM activities (generation, classification, optimization, prediction) were mapped to telecommunication applications. Such applications include network issue troubleshooting, network defect detection, and traffic load level prediction.

Blockchain: Geren et al. [85] surveyed how blockchain techniques can support the security and safety of LLMs, for example by verifying training data authenticity and privacy preservation. On the other hand, He et al. [105] surveyed LLMs for supporting blockchain security. They showed that LLMs can support the blockchain community by detecting vulnerabilities in the source code of smart contracts, detecting irregular transaction patterns or generating of smart contracts.

Robotics: Kim et al. [146] explored the use of LLMs in robotics. Their focus is on recent LLMs (after GPT-3.5) and text-based LLMs, while still allowing the inclusion of relevant multimodal approaches. They distinguish four main categories of LLM use: communication, perception, planning, and control. Additionally, they provided guidelines for prompting LLMs for four robotic tasks: interactive grounding, scene-graph generation, few-shot planning, reward function generation.

Similarly, the survey by Zeng et al. [389] presented LLM applications in robotics with regards to control, perception, decision-making and path planning. Different from Kim et al. [146], they put more emphasis on LLMs and transformer architectures, as well as challenges. Lastly, Shi et al. [276] addressed the use of LLMs in socially assistive robots (SARs) with a short survey. Herein, challenges and opportunities of using LLMs in SAR were discussed.

Hardware: Similar to their use in detecting software vulnerabilities (Section 6.2), LLMs can support the security of hardware components. Makhzan and Kamali [210] compared 10 such studies.

Agriculture: Zhu et al. [429] reviewed how LLMs and vision models can be applied in agriculture.

7 Multimodality

Generally, LLMs are applied to textual data and excel at language-based tasks. However, their application has been extended beyond texts to further modalities, for which we discuss relevant surveys. Wu et al. [337] outlined the history of multimodal approaches, from single modality to recent large-scale multimodal systems. Yin et al. [382] presented information on the architecture of Multimodal Large Language Models (MLLMs) as well as their training and evaluation. Song et

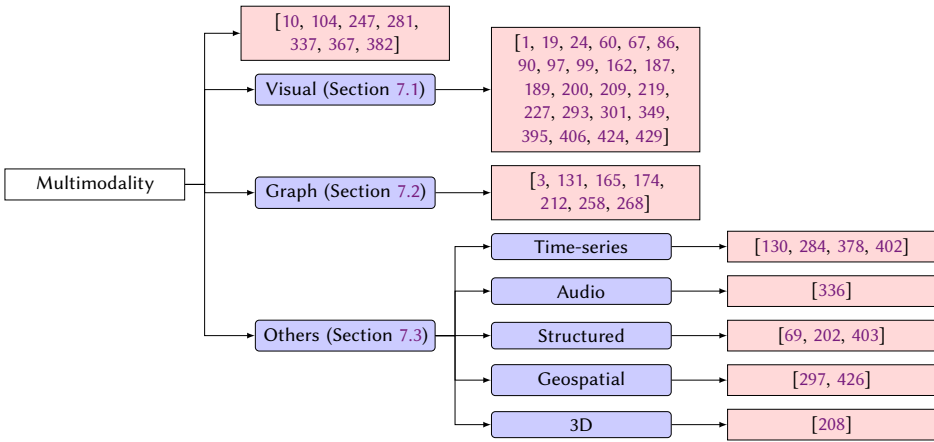


Fig. 7. Taxonomy of surveys on multimodality.

al. [281] described how different modalities can be aligned. Other surveys studied the generation and editing across modalities [104], training data [10, 247], or analysis of sentiments [367].

7.1 Visual

The most frequent application of multimodal language models we found is for visual tasks, with multiple surveys presenting comprehensive overviews [19, 395]. For example, Zhang et al. [395] gave background information on the visual paradigm as well as a summary of characteristics such as downstream tasks of Vision Language Models (VLMs) and their architecture. Among others, they outlined datasets and pre-training methods. There are several other comprehensive surveys, which put different foci, such as datasets [97], models [86], or details on regular LLMs [24].

The surveys by Du et al. [60] and Long et al. [200] focused on pre-trained vision-language models. First, data is transformed into desired representations. Afterwards, an architecture is designed to model the interaction between text and image. Further surveys took prompting [90], fine-tuning [349], and the detection of out-of-distribution samples and anomalies [219] into account. These advancements enabled the application of VLMs in diverse domains with surveys describing their applications in agriculture [429], medicine [99], autonomous navigation [209, 406], document understanding [1], and video analysis [227, 293, 424].

While VLMs offer advantages in several tasks, they can be vulnerable to attacks, which affects their usability in real-world applications [67]. Here, Liu et al. [187] surveyed four types of attack methods (adversarial attacks, jailbreak, prompt injection, and data poisoning) as well as potential defense methods. Fan et al. [67] considered different attack scenarios based on the type of model access (i.e., white-box, gray-box, black-box). Ethical AI has been further taken into account by Vatsa et al. [301] who surveyed bias, robustness, and interpretability of VLMs. Lee et al. [162] solely focused on biases and their mitigation. Another shortcoming of VLMs are hallucinations, which was surveyed by Liu et al. [189]. They collected methods and benchmarks for evaluating hallucinations and mitigate them. In total, there are five areas that have been addressed for mitigation: data, vision encoder, connection module, LLM, post-processing.

7.2 Graph

Jin et al. [131] created a comprehensive survey on different ways LLMs can interact with the structured information provided in graphs. Hereby, there are three types of graphs to consider: pure

graphs, text-attributed graphs (e.g., nodes have texts), text-paired graphs (a complete graph is paired with text). Additionally, LLMs can be used in three different manners for graph tasks: as predictors, encoders (e.g., encoding node texts as vectors), or for aligning text encoding with Graph Neural Networks (GNNs). Their taxonomy considered the intersection of two dimensions: the application scenario (graph type) and the LLM technique. Moreover, they created an overview of datasets from different domains (e.g., academia, e-commerce, books, Wikipedia) and graph problems studied (e.g., shortest path, neighbor detection).

Other taxonomies are included in the works by Ren et al. [258] and Li et al. [174]. The taxonomy by Ren et al. [258] considered four aspects: GNNs as Prefix, LLMs as Prefix, LLMs-Graphs Integration, LLMs-Only. For GNN as prefix, data is first processed by GNNs and then fed into LLMs. LLMs as Prefix does the opposite, processing data with LLMs to improve GNNs. LLMs-Graphs Integration entails methods that improve the ability of LLMs to handle graph data, while LLMs-only describes work that applies LLMs to graph tasks via prompting. Li et al. [174] devised a taxonomy with three categories: enhancer (enhancing quality of node embeddings), predictor (using LLMs for prediction in graph-tasks), and alignment (aligning embedding spaces of LLMs and Graph Neural Networks).

Mao et al. [212] studied the integration of LLMs for Graph Representation Learning (GRL). They outlined existing approaches for using LLMs to improve GRL tasks. Approaches are investigated with regards to four components: knowledge extraction, knowledge organization, integration strategies, and training strategies. Shang and Huang [268] surveyed the use of LLMs for graph analytics tasks. Their survey considered three aspects, the processing of graph queries with LLMs, inference and learning over graphs, and applications. Ample visual examples are provided for the tasks (graph understanding, graph learning, graph-formed reasoning) and prompts.

While the outlined surveys addressed graphs in general, we found two surveys focused on **knowledge graphs**. Such graphs are used to model and structure knowledge bases. On one hand, Agrawal et al. [3] surveyed how knowledge graphs have been used to combat hallucinations in LLMs. For this, they defined three groups: inference (e.g., RAG), training (e.g., pre-training, fine-tuning), and validation (e.g., fact-checking LLMs). On the other hand, Li and Xu [165] addressed both, how LLMs can enhance knowledge graphs and how knowledge graphs can enhance LLMs.

7.3 Others

Beyond the extensively researched domains of vision and graphs, MLLMs are expanding to a broader range of formats. The following surveys cover these emerging modalities, each presenting unique challenges and opportunities when integrated with LLMs.

Time-series: Jiang et al. [130] created a survey on time-series analysis with LLMs. LLMs can model time-series via querying, tokenization, prompting, fine-tuning, or the integration of LLM output in existing models. Overall, this survey includes 21 studies, over various applications (e.g., CV, mobility, healthcare, finance), for which the modeling approaches, tasks, and underlying LLMs are extracted. The survey by Ye et al. [378] contains time-series studies for similar application domains, however, their analysis focused on three dimensions: effectiveness, efficiency, and explainability. Su et al. [284] included a discussion of anomaly detection for time series.

Beyond this scope, Zhang et al. [402] considered visual representations of time series as well as LLM-based tools to support the processing of time-series, for example by creating code.

Audio: By converting audio into discrete codes, they can be processed by language models. Wu et al. [336] provided an overview of six neural models and 11 language models for processing audio. For each language model, they presented the addressed tasks as well as input and output format.

Structured: Tables represent data in a structured, two-dimensional manner and can be processed with LLMs. Fang et al. [69] reviewed techniques, metrics, datasets, and models for four techniques for applying LLMs to tables: serialization, table manipulation, prompt engineering, and end-to-end

systems. Emphasis is also put on the use of LLMs to generate tabular data. In addition to discussing training approaches for LLMs and Visual language models, Lu et al. [202] described prompting techniques and the use of agents.

Zhang et al. [403] focused their survey on techniques to improve the performance of LLMs for different table processing tasks (QA, fact verification, table to text, text to SQL). For five popular improvement techniques, they showed a performance comparison over four datasets.

Geospatial: Zhou et al. [426] surveyed LLMs with geo-perceptive capabilities to handle multiple modalities of geospatial data. They focused on a specific family of language models, Vision-language geo-foundation models (VLGFM). These VLGFM incorporate diverse data modalities (satellite images, geo-tagged text, remote sensing images) to address a wide range of geospatial tasks (e.g., image captioning, visual grounding). The survey includes an overview of tasks, datasets and metrics for evaluation as well as a description of model architectures. Tucker [297] reviewed LLMs for Geospatial Location Embeddings (GLE) to represent and express space.

3D: LLMs have seen use in spatial tasks, which require the consideration of three dimensions. In particular, Ma et al. [208] investigated how LLMs can understand and interact with 3D data. Their survey provided information on different 3D data representations (e.g., point cloud, grid, mesh), tasks (captioning, grounding, conversation (question answering), agent, generation), and datasets. Additionally, the LLMs and 3D components for 37 publications are extracted and described.

8 Risks and Mitigation

While prior sections outlined the benefits in various domains, LLMs can be susceptible to bias and safety issues or share private information [42, 197]. These concerns propagate to various fields [47], such as healthcare [96] or education [364], and are major challenges that need to be overcome to achieve trust [71, 119, 184, 197, 301] and transparency (e.g., by explaining responses) [20, 203, 253, 413]. Researchers showed interest in the different types of risks and their mitigation [266].

In the following, we discuss the main concerns pointed out and covered by existing surveys: fairness, hallucinations, security, and privacy [40, 56, 81, 119, 144, 154].

8.1 Fairness and Bias

LLMs can propagate social biases from the training data, causing fairness and bias issues, which has been covered by several studies [39, 75, 172]. The survey by Gallegos et al. [75] is the most comprehensive with three taxonomies, one for metrics, datasets, and bias mitigation methods each. Chu et al. [39] presented toolkits in addition to datasets, while Li et al. [172] took model size into account. They distinguished fairness studies based on LLM size, as smaller models allow for fine-tuning, while large models are prompted instead.

Surveys have also focused on a specific aspect, such as metrics [50] or the debiasing of LLMs [184]. Another set of works surveyed specific fields for biases, such as education [160], e-commerce [257], information retrieval [44], vision-language models [162], or recommender systems [263].

Lastly, Wang et al. [314] collected human perspectives on LLM bias from several studies and summarized their perspectives. Among other things, people perceived bias more when they failed to receive desired responses.

8.2 Hallucination

At times, the outputs generated by LLMs are inconsistent with the actual answer or the user input itself, which is called “hallucination”. There are three comprehensive surveys addressing this issue [116, 377, 405]. They contain details on causes, benchmarks, and mitigation approaches. We have also found two surveys addressing hallucinations for vision-language models [11, 189].

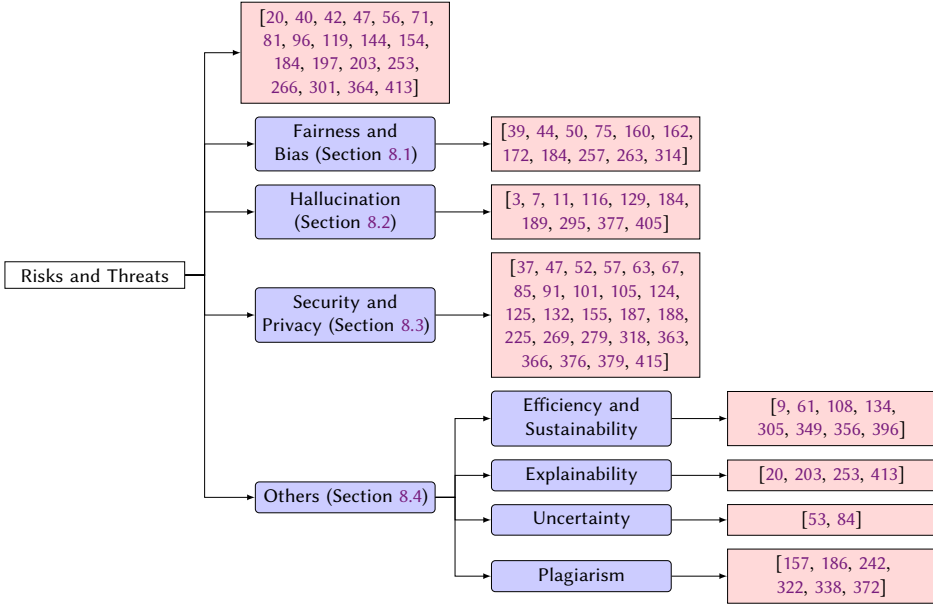


Fig. 8. Taxonomy of surveys on risks and threats.

As hallucinations can be detrimental to LLM usage, their mitigation (“dehallucination”) is studied. We encountered surveys that either gave an overview of mitigation techniques [184, 295], or elaborated on a single approach in detail, such as knowledge graphs [3] or knowledge augmentation [7].

A different point of view is provided by the survey of Jiang et al. [129], who considered hallucinations as a chance to foster creativity in LLMs.

8.3 Security and Privacy

Yao et al. [376] conducted a holistic survey on the security and privacy of LLMs, including the use of LLMs to support privacy by, for example, generating secure code or testing applications. Security has further been considered for multimodal LLMs [67, 132, 187, 269], agents [101, 318], and blockchain [85, 105]. In addition, Derner et al. [52] considered the use of LLMs for malicious intents, such as writing spam messages or sharing fake news.

Remaining surveys consider the security and safety risks of LLMs themselves. Particularly, security and privacy issues can be exploited passively by leaking personally identifiable information via regular interactions with a model, or dedicated attacks [47, 279, 363]. In terms of attacks, existing surveys have summarized them into the categories listed in Table 4.

The majority of these surveys not only describe the attacks, but also provide details on their mitigation. While attacks are usually automatic, some surveys list methods with human interference [37, 57, 269]. Attacks can be carried out on the provided service, by assessing or changing the prompt or responses, or slowing the model down with intensive prompts [52, 318]. Likewise, attacks can be performed by the service provider, for instance to recompute user queries [91].

8.4 Others

In addition to core risks detailed previously, a holistic assessment requires considering broader challenges related to their development and use in society. The following surveys address several of these considerations.

Table 4. Overview of attack techniques and surveys which describe them.

Security Attacks	Backdoor/data poisoning attacks [37, 47, 52, 57, 63, 124, 187, 188, 318, 363, 366, 415]
	Jailbreak [37, 47, 132, 187, 188, 269, 318, 379]
	Adversarial [37, 57, 91, 187, 269, 318]
	Prompt injection [37, 47, 52, 57, 63, 155, 187, 188, 269, 318]
Privacy Attacks - Data	Membership inference [47, 52, 63, 125, 225, 279, 318, 363]
	Attribute inference [91, 225, 279, 318, 363]
	Corpus inference [63, 91, 318, 363]
	Data extraction attack [47, 225, 279, 318]
	Knowledge stealing attacks (With RAG) [318]
Privacy Attacks - Model	Model extraction/stealing [52, 63, 318, 363]
	Accuracy extraction [91]
	Fidelity extraction [91]

Efficiency and Sustainability: The training and fine-tuning are compute-intensive activities that require time, compute resources, and energy, which introduces a critical challenge.

Efficiency concerns have been addressed for both text-based language models and multimodal language models [134, 349, 356]. For instance, Xu et al.’s [356] focus was on resource-efficiency, which entails the improvement of the efficiency of LLMs but also making better use of the available resources (e.g., data). Thereby, some of the methods would increase performance and not necessarily efficiency (e.g., better performance via data augmentation). Bai et al. [9] categorized resource efficiency methods based on the resources they optimized: computational, memory, energy, financial, and network. Wan et al. [305] listed frameworks which provide efficiency methods. Duan et al. [61] did not only consider efficiency but also fault tolerance. This is important to ensure that the computations are reliable and do not get lost. Further efficiency considerations for the different components of LLMs can be found in Section 4.

In terms of sustainability, Hort et al. [108] considered the sustainability of training and fine-tuning LLMs, as well as sharing them. By making trained LLMs publicly available, users do not need to train them themselves and save energy. Zhang and Chen [396] presented how LLMs can support in improving the energy efficiency of buildings and aid in decarbonization activities.

Explainability: In the context of LLMs, explainability refers to methods used to understand the reasoning behind their generation. These models are often categorized as “black-box” systems since their inner reasoning is challenging to interpret. Explainability allows to understand the model’s behavior, which increases user trust, allows debugging, and helps guide their development.

Some authors provide a more general overview of the field. Zhao et al. [413] proposed a taxonomy classifying these methods based on two training paradigms: traditional fine-tuning-based and prompting-based. For each paradigm, they further discussed the goals to achieve local and global explanations. Luo and Specia [203] classified methods into local analysis and global analysis. Local analysis includes feature attribution and transformer block analysis, while global analysis focuses on mechanistic interpretability and probing-based methods. Meanwhile, Cambria et al. [20] explored the balance between interpretability and functional advancements. They labeled each work based on one or more primary objectives: comparison of models, explanation, improvement, interpretability, and reasoning. However, a common thread in these overviews is the challenges of evaluating explainability [203, 413]. In this landscape, Rai et al. [253] provided insights and challenges in the subfield of mechanistic interpretability which focuses on reverse engineering operations to understand LLMs. This approach analyzes features and circuits of a model to provide insights and enable practical applications in fields such as knowledge editing and AI safety.

Uncertainty: To determine the confidence of LLMs with their generated responses, their uncertainty can be determined [53, 84]. This metric can indicate the likelihood that a response is correct, which is critical for establishing model reliability.

Plagiarism: The content generated by LLMs can also be seen as a potential threat, as one could misuse them for plagiarism [242]. Therefore, other studies set out to detect LLM-generated content [338, 372]. Another approach to identify as well as protect the ownership of LLM generated texts is watermarking [157, 186, 322].

9 Conclusion

Many secondary reviews have been written about LLMs. We have reached a point where the same surveys that set out to bring order and structure now contribute to the convolution of the space. To address this challenge, we performed a review of the tertiary literature to gather and analyze LLM-related surveys, reviews, and mapping studies. We have identified 984 secondary studies through our systematic search, of which 424 are included in this article (all studies can be found in our GitHub repository and replication package). We devised a taxonomy and categorized surveys by their main focus (e.g., fine-tuning of LLMs, application to software engineering tasks). By doing so, we aim to provide researchers with a useful resource for navigating the growing field of surveys, to which we are adding another one.

Data Availability

To support research in the area and enable replication and verification of our work, a replication package containing the complete collection of studies analyzed in this research, together with a bibliography file and all associated metadata tags, is publicly available on Zenodo.⁵

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